

Towards Monitoring and Analytics for Sports Exercises

Submission for ST Microelectronics AIoT Craft Competition

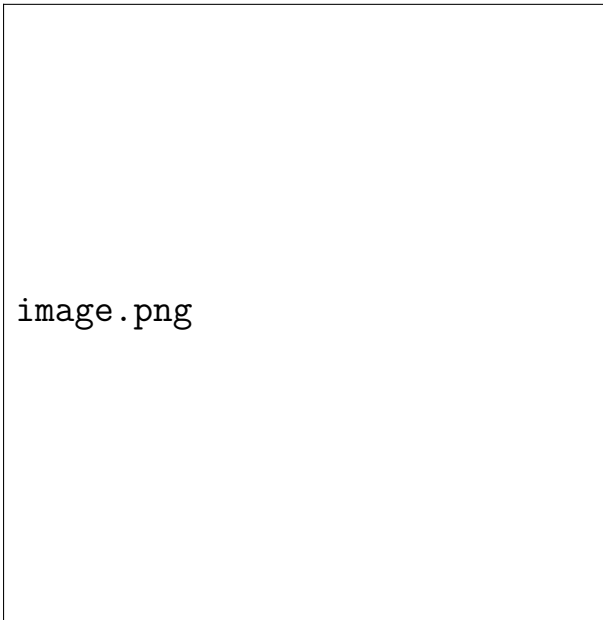


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1 Problem Description

Specialized exercises are crucial for success in competitive sports. Athletes perform exercises under the guidance of a coach or physical trainer. In existing methods, the physical trainer picks up visual cues to improve the quality of training. In specialized cases with necessary financial support, key-point detection-based methods are used which leverage computer vision techniques. None of these methods use advanced data analytics.

Advanced data analytics using machine learning can be used to significantly enhance the training regimen and outcome. Even non-athletes can use such methods to understand their exercise regimen. Not only can the proposed method significantly improve exercise training for athletes, but it can also democratize access to advanced analytics for physical exercise in general.

The STEVAL-MKBOXPRO[1] is a wireless kit that enables IoT-based applications through remote data gathering and evaluation. It comprises an accelerometer, gyroscope, temperature, pressure sensors, and a microphone. ST AIoT Craft is an online tool that programs ST sensors—equipped with a machine learning core (MLC)—to run inference operations. This project aims to demonstrate a proof-of-concept for advanced analytics in sports training, specifically in gym exercises using wearable sensor platforms.

2 Research Methodology

To develop a solution for monitoring and analyzing gym exercises, our team conducted an extensive review of both academic and commercial approaches, and researched mobile applications that track users' movements through sensors. In our academic research, we studied papers in the field, including one on sensor-based exercise recognition using wearable sensors and LSTM neural networks [2] and another that proposes a wearable sensor-based exercise monitoring system employing a Multi-Attribute Fuzzy Evaluation Model [3]. These studies gave our team the knowledge required to tackle the problem at hand, including how we could approach the core problem of data acquisition.

In parallel, we evaluated several commercial mobile applications designed to track physical activities such as gym exercises, walking, jogging, and running. Although these apps offer basic tracking features using sensors on the mobile phones, none of these applications integrated wearable sensor data recording with real-time analytics and result logging.

To summarize, what we learned from our research is that the academic studies provide methodologies to train models for predicting whether the exercises are being done correctly by the user, while commercial apps focus on general movement tracking. There is no wearable sensor-based mobile application that both records detailed sensor data and delivers in-depth analytics. This gap could effectively be addressed by the STEVAL-MKBOXPRO, the ST AIoT Craft website and mobile application.

3 Role of ST AIoT Craft

The ST AIoT Craft platform [4] plays a pivotal role in our solution by serving as the bridge between the captured sensor data and real-time analytics. This platform integrates

the data recorded through multiple sensor inputs and can train Decision Tree models based on the recorded data and labels provided by the user. Also, through its built-in machine learning core, the ST AIoT Craft app not only captures raw sensor data but also is capable of processing it on the fly, allowing athletes, coaches and fitness enthusiasts to gain real-time insights.

The website provides an interface for the user to develop their own project and machine learning decision tree classifier, while the mobile application leverages the full features of the board and allows the user to log the data recorded by the sensors, as well as validate the model in real-time based on the training data supplied by the user.

In our project, we used the ST AIoT Craft web application to create our datasets for logging data, creating our own custom project and using the datasets to train the Decision Tree model. We used the ST AIoT Craft mobile application to log the data through the STEVAL-MKBOXPRO into the created datasets, and to use the trained classifier for real-time classification to validate our trained model. These steps will be elaborated further in the upcoming sections.

4 Proposed Solution

This section details the steps taken by our team in building the proposed solution. We have incrementally built two versions of our solution, which we call *version 1* and *version 2*. We will first describe version 1, which involves the ST AIoT Craft tool suite, explain our reasons for the creation of version 2 and then finally describe version 2.

4.1 Version 1

Data Acquisition

For the purpose of demonstration, we selected five gym exercises, specifically those involving the arms and those not involving the simultaneous use of both hands. These exercises were:

1. Bicep Curls
2. Lateral Raises
3. Side-arm extensions
4. Tricep curls
5. Hammer curls

Each unit of an exercise is called a *rep*, and a *set* consists of a fixed number of reps of the particular exercise. Generally, an athlete would perform a set twice, once for each arm or leg. While performing the gym exercises, the subject logs data on the STEVAL-MKBOXPRO sensors. The relevant sensors in the scope of this project include the accelerometer and gyroscope [LSM6DSV16X] sensor readings.

In our proposed solution, the output of the decision tree classifier would either be '*proper form*', or '*improper form*', indicating whether the subject would be performing the exercise correctly or incorrectly, respectively. We created a dataset on the ST AIoT Craft website for each exercise and recorded the data for 3 sets of 15 reps each for these five exercises

from a professional gym instructor, which would indicate the correct way of performing the exercise. We attached the STEVAL-MKBOXPRO to the arm by fastening it using adhesive tapes, at a location where the movements of the target muscle could be detected. We call the dataset recorded by gym instructor as the 'golden data', and is labeled as 'proper form'. The golden data would be useful in the training phase of the decision tree. In parallel, we logged data for the 'improper form' label by performing the exercise in an incorrect manner for the same number of sets and reps.

Project Creation

The acquired data was present on the SD card of the STEVAL-MKBOXPRO. We uploaded the SD card data into our created dataset on the ST AIoT Craft website. We were able to verify the labels of the data and visualize the data graphs, as visible in Figure 1. Once the data was uploaded, we created our own project for each exercise by selecting the dataset we had created for the relevant exercise, which would be necessary to train the model.

Model Creation and Training

Now that we had labeled data, we could train our decision tree model from the ST AIoT Craft website. We trained the models through the *expert* mode present in ST AIoT Craft, and obtained a classification accuracy of 100 percent for all exercises, as shown in Figure 2 for a sample exercise. We shall now detail on the parameters used in the expert mode. Specifically, we used the *automatic parameter extraction* feature provided in the website. We extracted all three parameters automatically - Window length, Filters and Features.

Filter and Feature Settings

The *Exhaustive search* option was used for the Filter settings, as this would select the optimal filters to be used by checking all possible combinations, leading to greater accuracy.

For the Feature settings we chose the following three extraction methods to be preferred:

1. Analysis of Variance (ANOVA)
2. Ada Boost (ADA)
3. Recursive Feature Elimination (RFE)

ANOVA was used to distinguish between sensor readings within an exercise. As our dataset samples were not distributed equally between the 'proper form' and 'improper form' labels, we chose the ADA model, as it is faster which works better than random forest extractors in these situations. Similarly, we chose RFE to eliminate redundant and irrelevant features and preferred it over Sequential Feature Extraction due to its faster speed. Our research for the feature extraction methods was based on [5], [6].

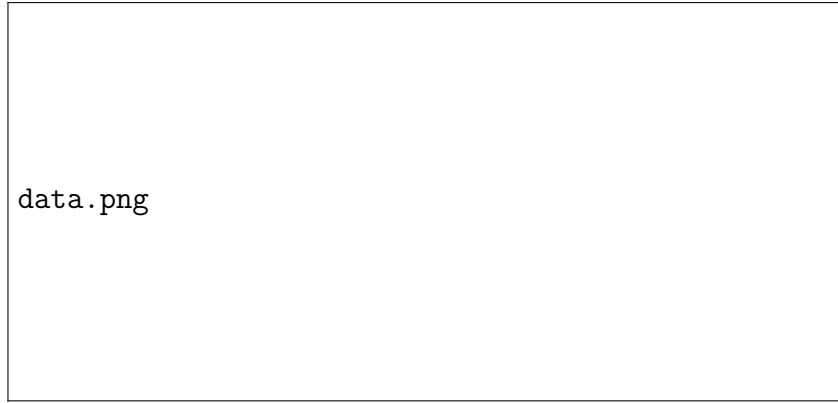


Figure 1: Sensor data

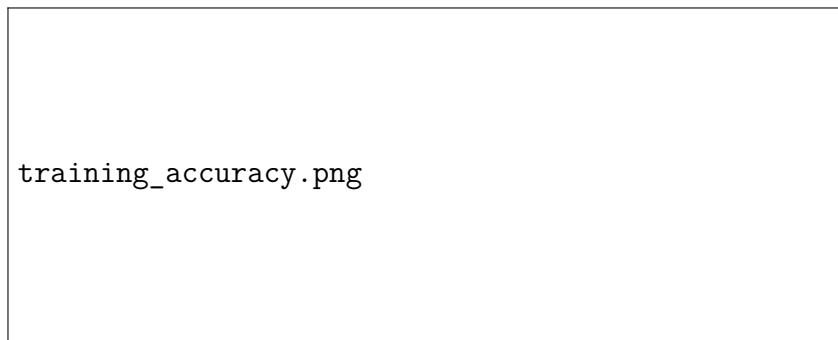


Figure 2: Model parameters and accuracy

Model validation on STEVAL-MKBOXPRO

Our next task was to validate the trained model using the ST AIoT Craft app against the subject's movements in real-time, and verify whether the model's classification output matched the movements made by the user. The relevant model for each exercise was downloaded on the STEVAL-MKBOXPRO board. For a sanity check, when the professional gym instructor performed the same exercise again, we observed that the real-time classification gave the output as 'proper-form', as we expected. Similarly, on making random movements, we got the output as 'improper form', thus showing that our model was working properly.

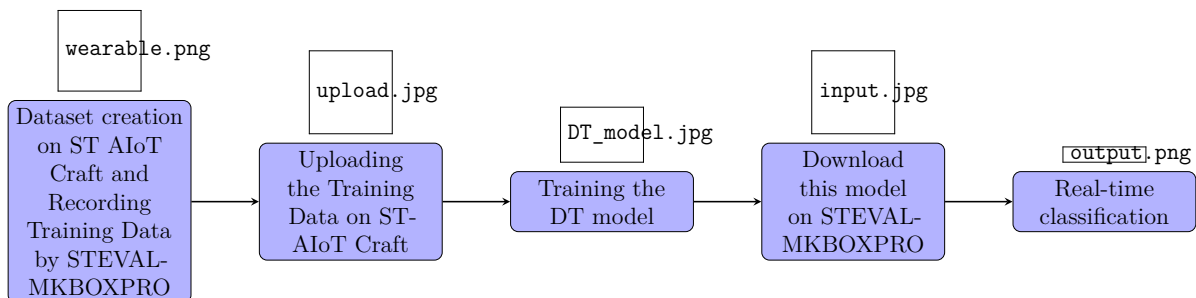


Figure 3: Stage 1 solution architecture

Figure 3 shows the organization of the steps in brief.

4.2 Version 2

Although the ST AIoT Craft suite that includes the website and mobile application tools is an acceptable solution for the problem at hand, it lacks features that can make the proposed solution of tracking gym exercises a more intuitive and easier experience for the end user. The main limitation we encountered was that data logging and validation cannot run simultaneously. Thus, the real-time data cannot be stored and used by the user at a later point in time. For example, the user would need the real-time data if they wanted to see how their performance varied across different exercise sessions, which would require data logs of the real-time classification. The data history of the user can be used for productive purposes, such as comparison of the user data against the golden data which would help the user to visualize their sensor readings against the expected readings and make adjustments to their form. To overcome these limitations and make the experience more user-friendly, our team developed a web application, *GymMetrics* that solved these problems. which would allow the user to upload the sensor data as a CSV file for the particular exercise for a particular date, and see how their exercise data compared against the golden data.

GymMetrics

GymMetrics has a simple structure in terms of organization of the website. It has a homepage, shown the Figure 4. Once logged in, Figure 5 is shown to the user, where the user may either view the past history and analytics of each exercise or, upload new data using the upload button. The golden data for each exercise is already pre-configured in the dashboard application.

Since simultaneous data logging and validation is not possible, the user would create a new dataset on the ST AIoT Craft website which corresponds to the raw sensor data of the user for each exercise. At the end of an exercise session, the user would have to download the CSV files from this dataset and this would be the data which the user would upload on the dashboard.

The upload page, shown Figure 6 requires the user to upload a CSV file for a particular exercise and for a particular date. The user can upload the downloaded CSV files from the ST AIoT Craft dataset. On successful upload, the user can then navigate to a particular exercise, which stores the history of the particular exercise, accuracy of the exercise performance and comparison against the golden data.

Since the exact parameters of the underlying decision tree model on the ST AIoT Craft website is unknown to us, we created our own decision tree model using the *DecisionTreeClassifier* model of the *scikit-learn* library in Python[7]. The accuracy graph is a history of the accuracy of the user. A sample accuracy graph is shown in Figure 7 For accuracy calculation, we used the following formula:

$$\text{Accuracy} = \frac{\sum_{i=1}^N y_i}{N} \times 100 \quad (1)$$

where:

- N is the total number of data points,
- y_i represents the classification of the i 'th data point (1 for correct, 0 for incorrect),
- $\sum_{i=1}^N y_i$ gives the total number of correct classifications.

As the recorded data consists of accelerometer and gyroscope data in the x , y , and z directions, we made 3-dimensional plots corresponding to the sensor data, and showed how their logs compared against the golden data. Figure 8 and Figure 9 show sample 3-dimensional accelerometer and gyroscope plots for test and golden data.

Hence, through the features available on the dashboard, the user can gain deeper insights on their exercise sessions over the ST AIoT Craft app which can be used to improve their performance. The Figure 10 shows the addition of the dashboard application in our solution architecture.

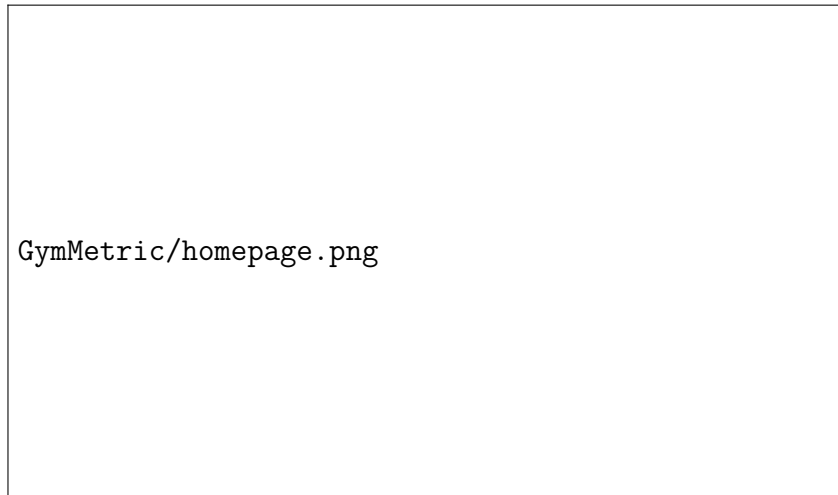


Figure 4: Home page of the Dashboard

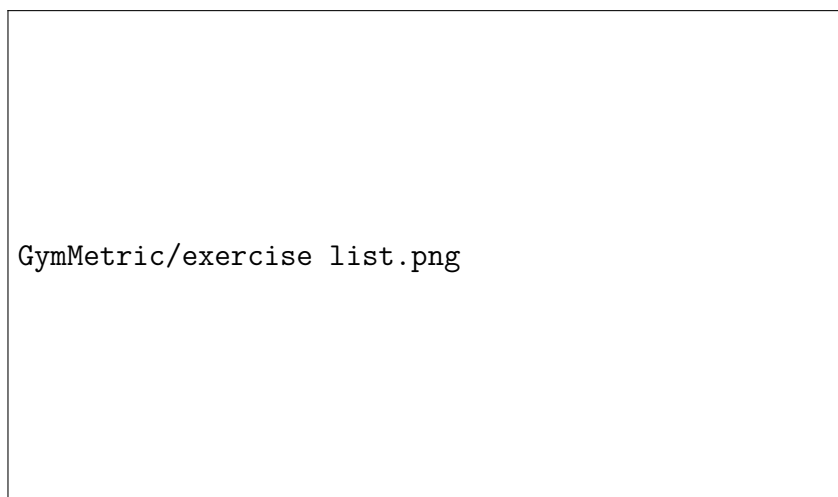


Figure 5: List of exercises

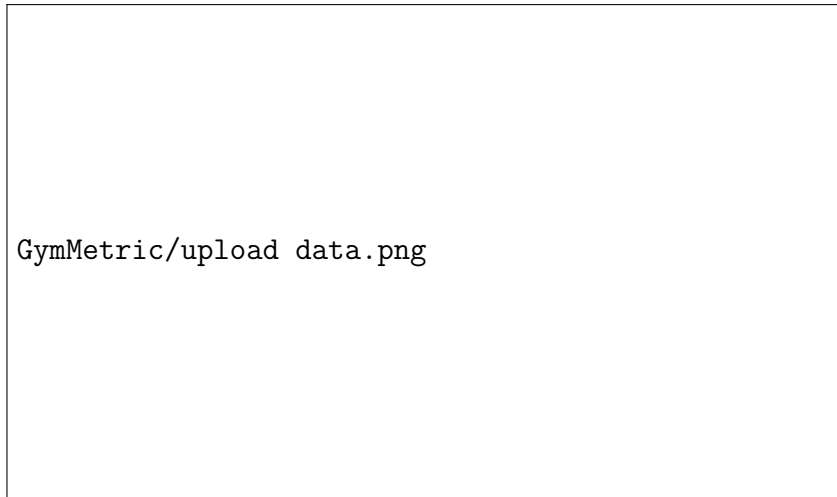


Figure 6: Data upload page

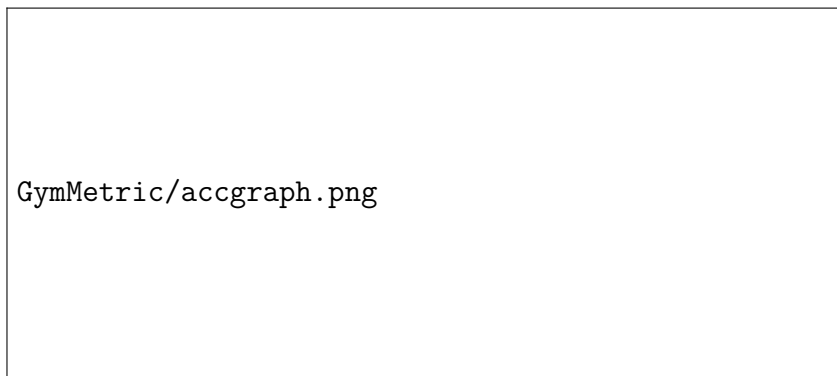


Figure 7: Accuracy Graph

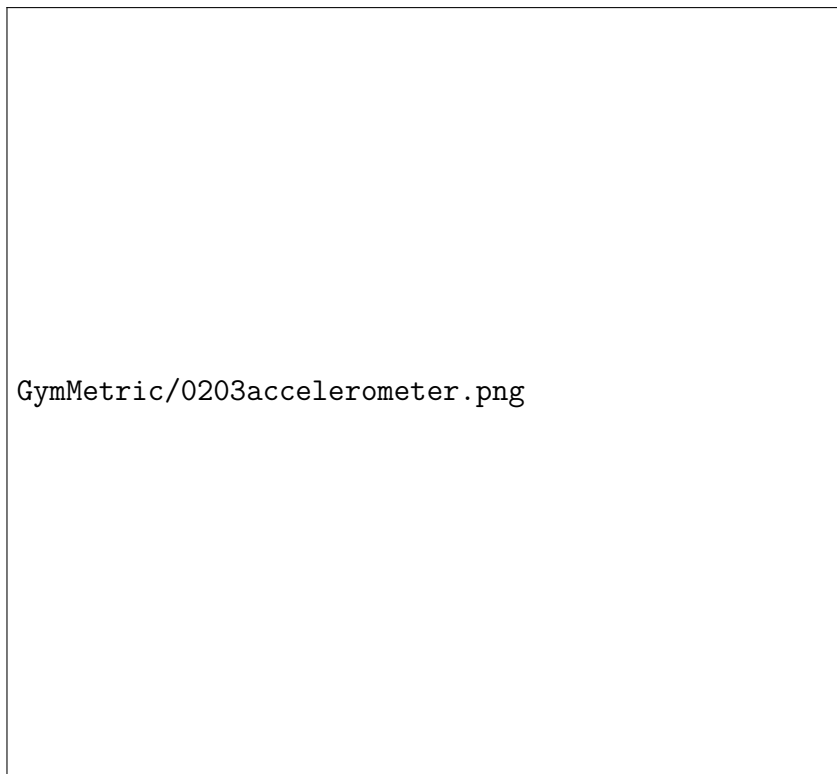


Figure 8: Accelerometer plot

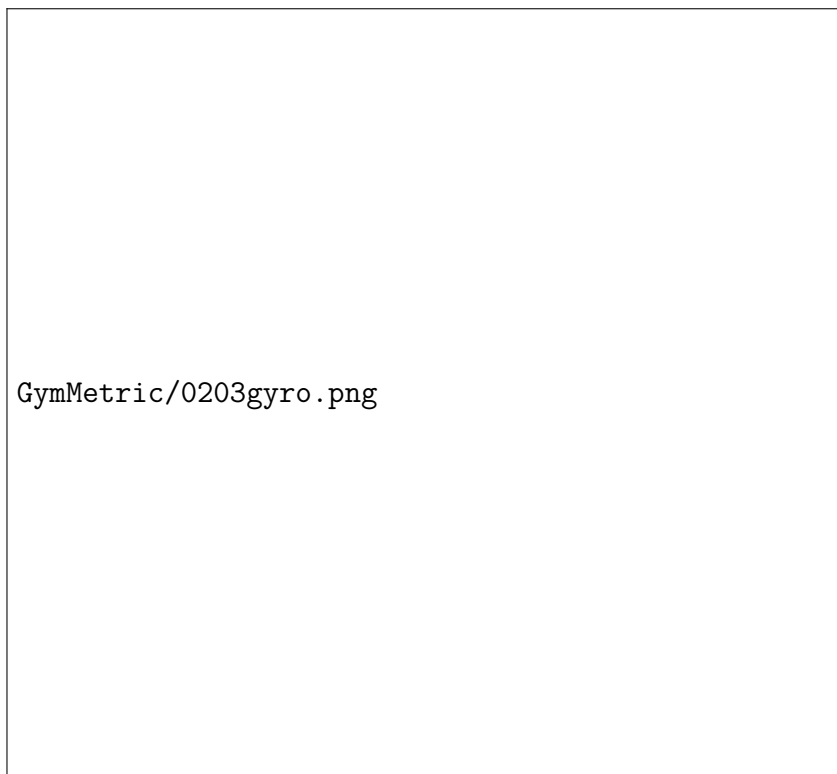


Figure 9: Gyroscope plot

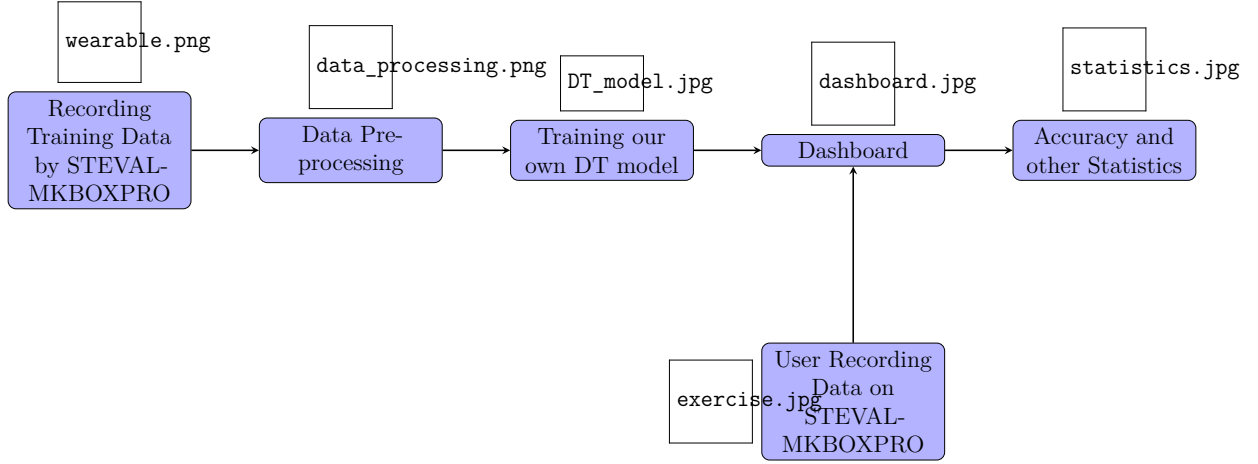


Figure 10: Stage 2 solution architecture

5 Conclusion and Future Work

Our current project has shown that the ST AIoT Craft tool suite, with the STEVAL-MKBOXPRO, can be used to successfully log accelerometer and gyroscope data, train the decision tree on the website, and validate the model in real time based on the trained data. This demonstrates the potential of integrating AI with IoT devices to provide intelligent feedback on user performance. To make the process more user-friendly and intuitive, our team also developed a web-based dashboard application where the user can upload recorded data logs to compare and analyze their performance against the pre-recorded golden data. This feature enables users to track their progress over time and identify areas for improvement. Our work is only a minor contribution, and while these results are promising, there is considerable scope for future work that can be incorporated to develop the idea into a complete solution for the end user. Expanding this system into a more comprehensive platform could significantly enhance the user experience by automating data collection, processing, and feedback mechanisms.

In the current implementation of the dashboard, the user has to manually upload the recorded logs by date. **Future iterations** of the system could enable seamless linkage between the user's ST AIoT Craft account and the dashboard account. This could be achieved through *API-based integration* with the ST AIoT Craft platform, allowing exercise data to be automatically uploaded immediately after a training session, eliminating the need for manual file uploads. Such integration would be crucial in turning this work into a *cloud-based physical fitness IoT platform*, where data is processed and stored centrally for better accessibility and analytics. Additionally, cloud storage would allow users to access historical data, track progress, and receive AI-driven insights from anywhere.

Other features could also be incorporated into the dashboard to make it into a commercially attractive product. For instance, it might incorporate features such '*family mode*' where one user may register their family members and a leaderboard metric could be developed where each member would be assigned a certain score based on exercise history, accuracy and additional parameters. A similar feature could be developed for friends.

Currently, users cannot log data when the model is being validated. The ST AIoT mobile

application could have an option to log data during the validation phase, and this data could be uploaded to the dashboard application. *API-based integration* could further enhance this process by enabling real-time synchronization between the mobile app, the dashboard, and cloud storage. This would ensure that users have continuous access to their exercise data, even during the validation phase.

The decision tree model supported by the ST AIoT Craft Suite is helpful for recognizing whether the exercise is being done correctly and performing basic classification. However, more complex models—such as regression models, deep neural networks, or ensemble methods—can be integrated to provide deeper insights into where the user’s form deviates from the golden data. By incorporating advanced machine learning techniques, the system could offer more personalized recommendations and detect subtle variations in movement that may impact performance or lead to injury. Furthermore, integrating real-time AI-driven suggestions could help users make immediate corrections.

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